What is this?

This notebook is part of a collection of `pluto` *notebooks on various topics discussed during the Time Domain Astrophysics course delivered by Stefano Covino at the [Università dell'Insubria](https://www.uninsubria.eu) in Como (Italy). Please direct questions and suggestions to stefano.covino@inaf.it.*

Table of Contents

A few important Fourier Transform

The rectangular function

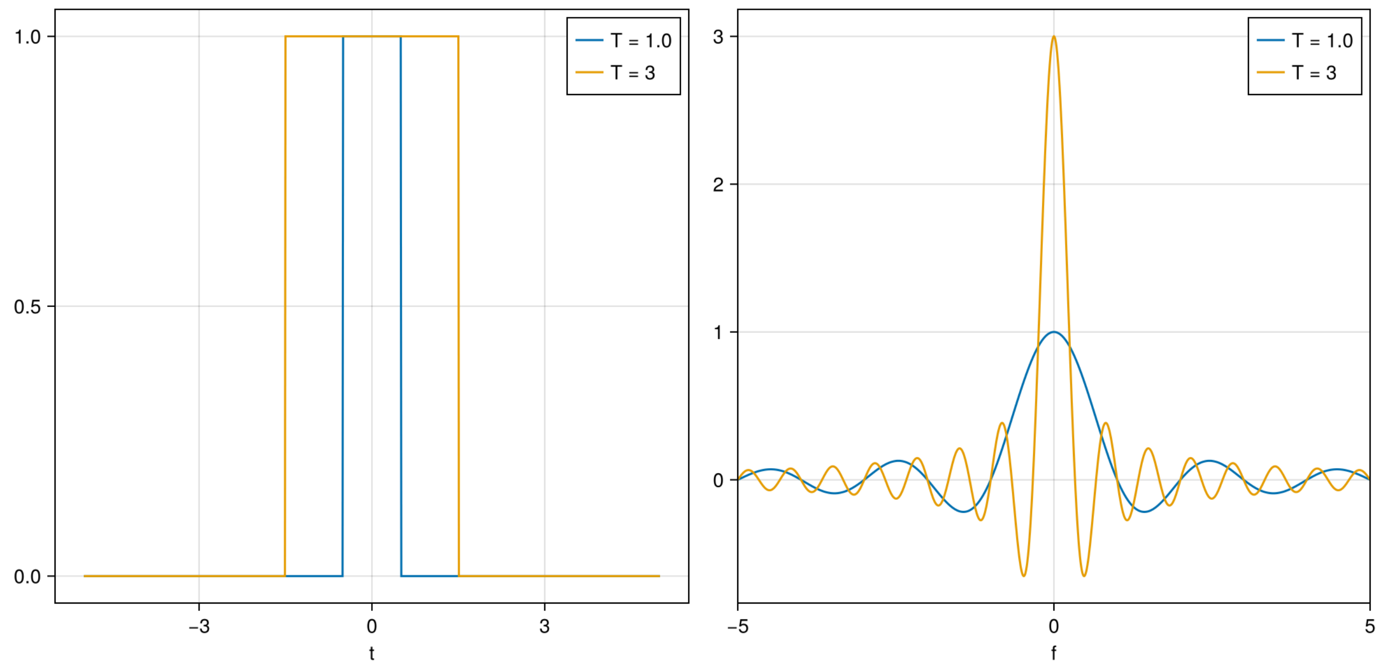
- We want to compute the FT of the rectangular function: e.g. a function $g(t)$ with a given amplitude and extending from $t = -T/2$ to $t = T/2$. For $|t| > T/2$, $g(t) = 0$.
- We will write the square pulse or box function as $rect_T(t)$, indicating that the rectangle function is equal to 1 for a period of T (from $-T/2$ to $+T/2$) and 0 elsewhere,

$$\mathfrak{F}\{g(t)\} = G(f) = \int_{-\infty}^{\infty} g(t)e^{-2\pi i f t}dt = \int_{-T/2}^{T/2} Ae^{-2\pi i f t}dt = \frac{A}{-2\pi i f}e^{-2\pi i f t}\Big|_{-T/2}^{T/2} = \frac{A}{-2\pi i f}[e^{-\pi$$

$$= \frac{AT}{\pi f T}\frac{e^{-\pi i f T} - e^{\pi i f T}}{2i} = \frac{AT}{\pi f T}\sin(\pi f T) \equiv AT\mathrm{sinc}(fT)$$

- where we have defined the "sinc" function defined as $\mathrm{sinc}(t) = \sin(\pi t)/\pi t$.

Width of the box function:



The Gaussian function

- We want now to compute the FT of the Gaussian function: $g(t) = e^{-\pi t^2}$.
- Let G(f) be the Fourier Transform of g(t), so that:

$$G(f) = \mathfrak{F}\{g(t)\} = \int_{-\infty}^{\infty} g(t)e^{-2\pi i f t}dt = \int_{-\infty}^{\infty} e^{-\pi t^2}e^{-2\pi i f t}dt$$

- Take the derivative of both sides of equation with respect to f, so that:

$$\frac{dG(f)}{dt} = \int_{-\infty}^{\infty} e^{-\pi t^2}(-2i\pi t)e^{-2\pi i f t}dt$$

- Writing $u = e^{-2\pi i f t} \rightarrow du = (-2\pi i f)e^{-2\pi i f t}dt$ and $v = ie^{-\pi t^2} \rightarrow dv = -2\pi i t e^{-\pi t^2}dt$, and given the formula for integration by parts: $\int udv = uv - \int vdu$ we have (the uv term becomes zero, because the limits are evaluated from $-\infty$ to ∞, where the product is zero):

$$\frac{dG(f)}{dt} = \int_{-\infty}^{\infty} ie^{-\pi t^2}(-2i\pi f)e^{-2\pi ift}dt = -2\pi f \int_{-\infty}^{\infty} e^{-\pi t^2}e^{-2\pi ift}dt = -2\pi fG(t)$$

- This is a first order simple differential equation for $G(f)$. The solution for this differential equation is given by:

$$G(f) = G(0)e^{-\pi f^2}$$

- All we need to do now to find $G(f)$ is figure out what $G(0)$ is.
- $G(0)$ is simply the average value of $g(t)$, because if you substitute $f = 0$ into the equation for $G(f)$ the complex exponential term goes away.
  - The integral has actually an elegant solution. The result is:

$$G(0) = \int_{-\infty}^{\infty} e^{-\pi t^2}dt = 1$$

- Therefore:

$$G(f) = \mathfrak{F}\{g(t)\} = e^{-\pi f^2}$$

Credits

This notebook contains material obtained by https://towardsdatascience.com/a-proof-of-the-central-limit-theorem-8be40324da83.

Course Flow

	Previous lecture	Next lecture	Course Summary
notebook	Spectral analysis	Spectral analysis	Course Summary
html	Spectral analysis	Spectral analysis	Course Summary

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